

Tokens & Tokenizers

Topics to discuss:

1. Why Tokens?
 2. What is a token?
 3. Types of tokens
 4. What is a tokenizer?
 5. The Full Flow
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1. Why tokens?

- Computers don't get words → only numbers
- so we convert: text → numbers

"I love AI" → [101, 2345, 789]

2. What is a token?

- A token = the smallest piece of a text that a model reads.
- Humans think in words. Models think in tokens.

"The cat is playing" → [The] [cat] [is] [play] [ing]

3. Types of tokens:

Word

"cat" → [cat]

Subword (★ Most common)

"playing" → [play] [ing]

Character

"cat" → [c] [a] [t]

Special

[CLS] start [SEP] separator
[PAD] padding

4. What is a Tokenizer?

Tokenizer → tool that chops text into tokens.
then give unique IDs to the tokens.

"I love ChatGPT" → [I] [love] [Chat] [GPT] → [101, 234, 890, 121]

5. The Full Flow:

Text → Tokenizer → Tokens → Token IDs → Model

6. Key idea:

- Same token → same ID. Always
- Every model would have its own tokenizer
- Token ≠ word. A token is often just a piece of a word.

Simple Analogy:

(*) Sentence → LEGO structure

(*) Token → LEGO blocks.

Next up : Hands-on : Tokens & Tokenizers